Justification: Why 2D Rotary Position Embedding?

We replace the original ViT's 1D learnable positional embedding with 2D Rotary Position Embedding (RoPE), adapted from Su et al. (2021) and used in vision models such as EVA-02 (Fang et al., 2023). Our rationale is based on four key differences:

1. Relative vs. Absolute Position Encoding
The original ViT learns a fixed vector for each patch position and adds it once at the input. The model must memorize absolute position-specific patterns. RoPE instead encodes relative positions by rotating the Query and Key vectors inside attention. The dot product between two rotated vectors depends only on the distance between their positions, not on absolute indices. This makes attention inherently translation-aware and should improve generalization, especially when training from scratch on limited data.

2. Explicit 2D Spatial Structure
The original ViT flattens the 2D patch grid (8×8 = 64 patches) into a 1D sequence and relies entirely on learned embeddings to recover spatial relationships. RoPE explicitly models 2D structure: the attention head dimension (64) is split in half — the first 32 dimensions encode row position and the second 32 dimensions encode column position. This provides the model with a strong 2D spatial inductive bias, which is particularly valuable for image data where horizontal and vertical relationships carry distinct semantic meaning.

3. Parameter Efficiency
The original positional embedding adds (num_patches + 1) × embed_dim = 65 × 192 = 12,480 learnable parameters. RoPE uses precomputed sinusoidal frequencies with zero additional learnable parameters. On a small dataset like CIFAR-100 (45,000 training images), reducing unnecessary parameters helps mitigate overfitting risk.

4. Per-Layer Positional Signal
The original ViT injects positional information once — at the input, before the first transformer layer. As information flows through 12 transformer layers, this positional signal may attenuate. RoPE applies position-dependent rotations to Q and K at every attention layer, providing a persistent and refreshed positional signal throughout the entire network depth.